What can waves tell us?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a signal's travel from the motion of its source.

SUCCESS CRITERIA - I can...

1. Identify what a wave can carry.
2. Separate signal travel time from source speed.
3. Name evidence needed to interpret an observation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A sphere's radius becomes 3 times as large at fixed density. Derive the area-to-mass multiplier. Explain why kinetic energy per mass alone does not predict survival during entry.

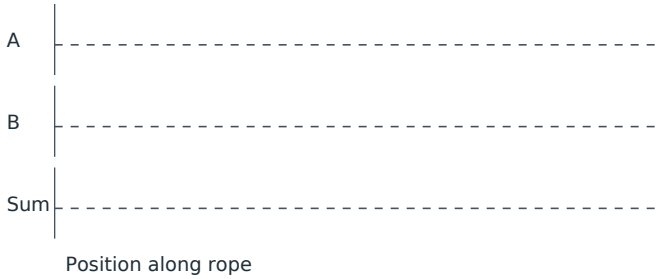
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: distance = speed × time
t = d/c for light in vacuum.
A signal takes time to travel. Receiving its light does not mean the source traveled with it.

Worked reasoning: A pulse travels 4 m along a rope in 2 s, so its average propagation speed is 2 m/s. A marked rope segment moves locally; it does not ride 4 m along with the pulse.

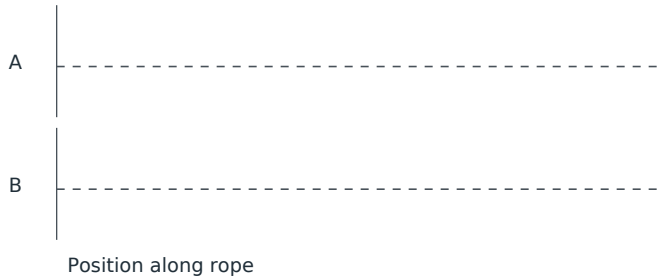
Your turn: Sketch three pulse positions and mark the same rope segment in each snapshot.



Marzano check - circle: ① ② ③

03 Read and interpret the model

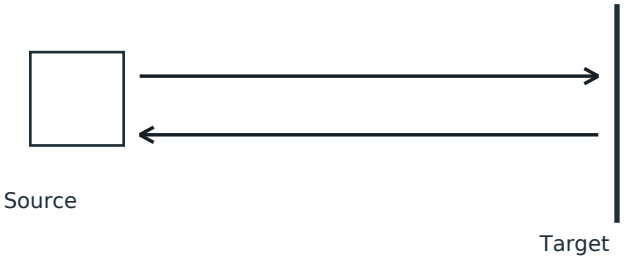
A pulse travels along a rope while each marked rope segment moves mainly up and down. Does the material travel all the way with the pulse?



Marzano check - circle: ① ② ③

04 Apply the physics

A radar station receives an echo 0.020 s after transmission. Use $c = 3.0 \times 10^8$ m/s. Label the outgoing and returning paths, calculate target range and explain the factor of 2.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student says seeing an asteroid’s light tells us exactly where it is “right now.” Explain one limitation.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Two objects at 290 K have different masses and materials. Must their internal energies be equal? Give two reasons. Label temperatures, samples and relevant energy transfers.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

07 Exit check

Two echoes return at 0.010 and 0.016 s with the same wave speed. Compare ranges and explain whether echo delay alone gives mass.



Marzano check - circle: ① ② ③

08 STRETCH: test the model

A signal travels 300 m in 0.9 s with timing bounds 0.85-0.95 s. Find central speed and bounds, treating distance as exact.

Distance (m)	Time (s)	Speed (m/s)
___	___	___
___	___	___

Marzano check - circle: ① ② ③